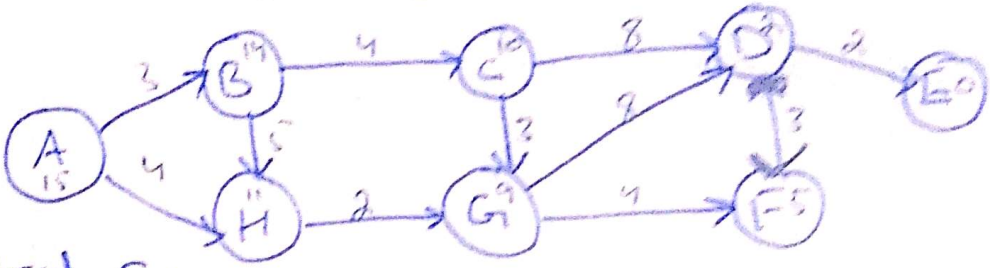
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L1F24BSCS0546

Artificial Intelligence

CP 2



→Best - First - Search

Step	Frontier	Expand	Explored
1	(A,15)	A	ϕ
2	(B,14), (H,11)	H	A
3	(B,14), (G,2)	G	A, H
4	(B,14), (D,2), (F,5)	D	A, H, G
5	(B,14), (F,5), (E,0),	E	A, H, G, D

Goal reached

Path = AHGDE

Path cost = 4 + 2 + 8 + 2 = 16

2) U C S

step	Frontier	Expand	Explored
1	(A, 0)	A	∅
2	(AB, 3), (AH, 4)	B	A
3	(AH, 4), (ABC, 7) ~~(ABH, 5)~~	H	A, B
4	(ABC, 7), (AHG, 6)	G	A, B, H
5	(ABC, 7), (AHGF, 10) (AHGD, 14)	C	A, B, H, G
6	(AHGF, 10), (AHGD, 14) ~~A B C D~~	F	A, B, H, G, C
7	~~(AHGD, 14)~~, (AHGFD, 13)	D	A, B, H, G, C, F
8	(AHGFDE, 15) Goal reached	E	A, B, H, G, C, F, D

Path : A → H → G → F → D → E

Cost : 4 + 2 + 4 + 3 + 2 = 15

3) A* Search

Step	Frontier	Expand	Explored
1	(A, 15)	A	ϕ
2	(H, 15), (B, 17) (A)	H	A
3	(AB, 17), (AHG, 15)	G	A, H
4	(AB, 17), (AHGF, 15) (AHGD, 16)	~~F~~	A, H, G
5	(~~AB~~, 17), (AHGD, 16) (AHGFD, 13)	D	A, H, G, F
6	(AB, 17), (AHGD, 16) (AHGFDE, 15)	E Goal	A, H, G, F, D

Goal reached

Path: A → H → G → F → D → E

Cost 15